Experimental and Model Characterization of Jet Fuel Emissions in a Micro-Turbojet Engine

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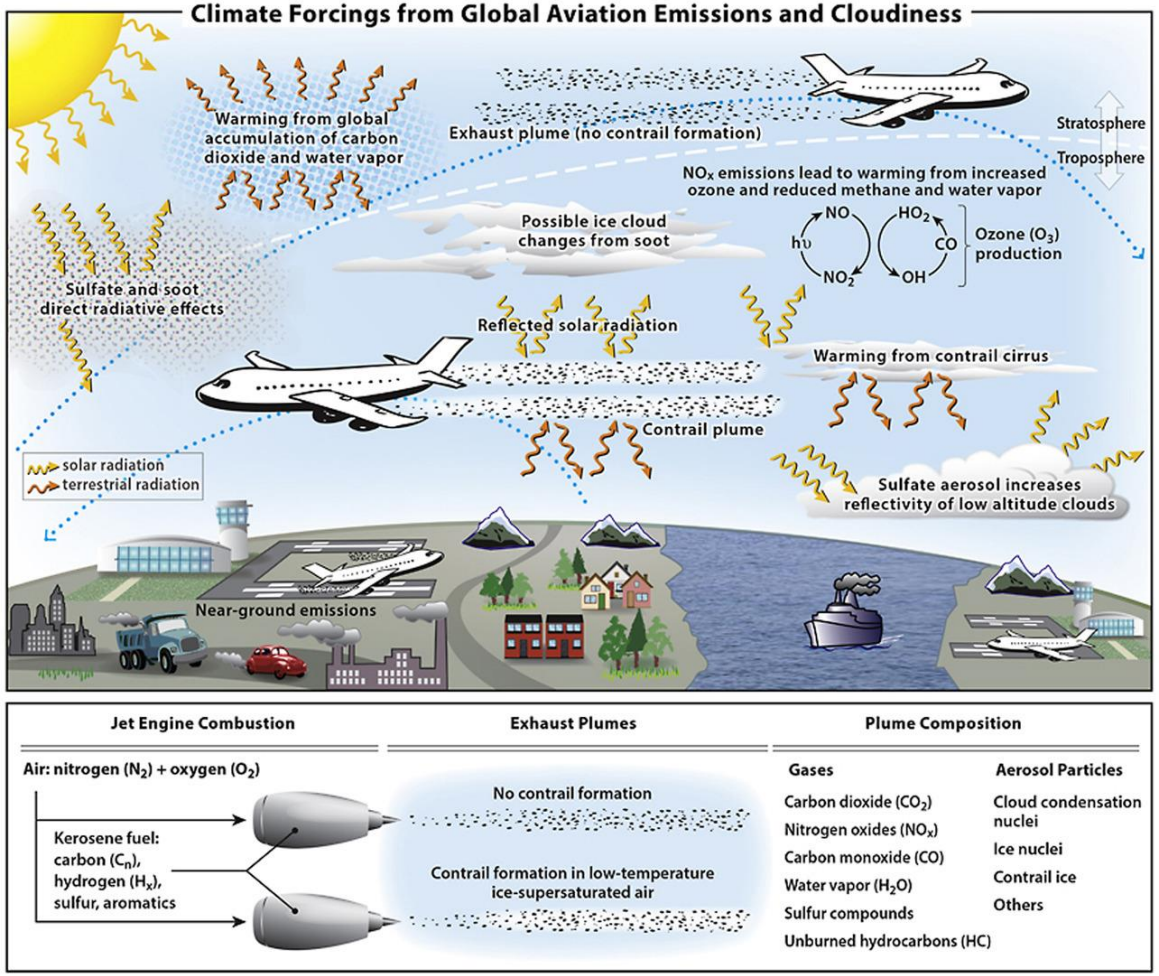
Aviation's Climate Impact

- Aviation contributes **3-5%** to global anthropogenic climate forcing [1,2].
- **More than half** of aviation's **total climate impact arises from non-CO_2 effects** [1].
- These effects are mediated by relatively small magnitude of emissions from aircraft engines that contribute disproportionately to radiative forcing [3,4]

Non-CO_2 Forcing Mechanisms

- Reactive nitrogen species drive near-field chemical transformations and interact with the atmosphere [5]
- Emitted soot can act as Ice Nucleating Particles (INPs), **forming contrails** & influencing the optical properties of clouds (albedo) which further amplifies radiative warming [3,7]

Because multiple coupled processes contribute to net climate forcing, CO_2 alone is an incomplete proxy for aviation's total impact.



Schematic overview of the processes by which aviation emissions and increased cirrus cloudiness affect the climate system. Net positive RF (warming) contributions arise from CO2, water vapor, NOx, and soot emissions, and from contrail cirrus (consisting of linear contrails and the cirrus cloudiness arising from them). Negative RF (cooling) contributions arise from sulfate aerosol production [1].

[1] (Lee et al. 2021), [2] (Gössling and Humpe 2020), [3] (Bickel et al. 2020), [4] (Teoh et al. 2022), [5] (Bräuer et al. 2021), [6] (Tadic et al. 2021), [7] (Gettelman and Chen 2013)

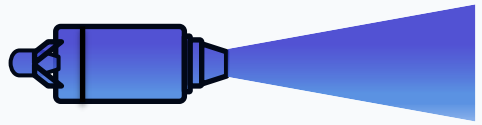
Industry Shift & Alternative Fuels

ICAO Regulations

ICAO has introduced standards for non-volatile PM (CAEP 11) and Annex 16 specifies limits for key gaseous non-CO_2 species [#].

The Interest in SAFs

Sustainable Aviation Fuels (SAFs) are positioned as a practical near-term pathway to reduce climate impact without altering existing airframes.



Emissions Data

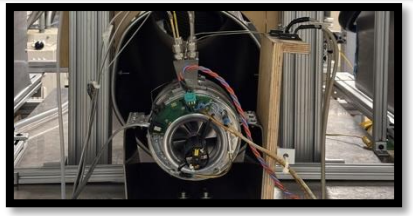
Conducting experimental work on full-scale turbofans to gather emissions data using SAFs is prohibitively expensive, logistically complex and tightly regulated by ASTM [#].

[8] (ICAO Annex 16)

MOTIVATION

Pollutant formation and resultant climate impact are highly sensitive to fuel chemistry, and operating point dependencies.
We need a controlled platform with a tractable physics informed approach to examine these sensitivities.

1. A Micro-Turbojet (MTJ) Test Rig



Why? Cost-effective, controlled platform
Enables repeatable experiments, (back-to-back tests) with modest fuel use across a wide operating envelope.

Micro-Max/Macro-Idle Intersection
At maximum thrust, the micro-turbojet reproduces the combustor inlet conditions of a full-scale turbofan at idle [#].

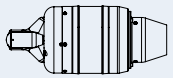
Combustor inlet conditions

	Micro-turbojet (Max Thrust)	Full-scale turbofan (Idle)
T_{t3}	469 K	453 K
p_{t3}	374.8 kPa	351 kPa
OPR	~4.0	~3.5

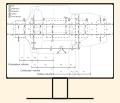
Intensive properties (T, p, ϕ) are very similar (scale-independent).

2. Couple with Physics-based Model

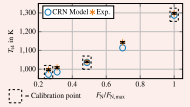
Physics-based 1D CRN model captures key chemistry and species formation.
Finite-rate chemistry with detailed kinetics is solved in idealized reactor elements that are arranged to represent the reaction zones of a RQL combustor [#].



Micro-Turbojet Experiment
Generate Jet A-1 data across multiple operating points



Performance Model
Links measured MTJ engine state to CRN input boundary conditions



CRN Model
Simulates rich–lean chemistry and species (CO_2, CO, NO_x, nvPM) formation

Powerful Together
Links measurable performance to chemistry, enabling interpretation and prediction beyond measured conditions.

3. What this Enables

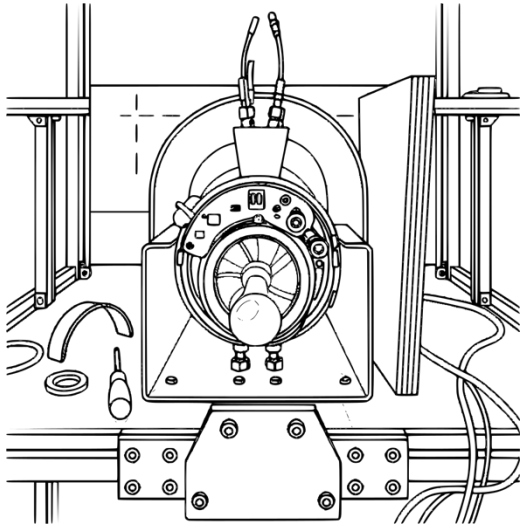
The coupled micro-turbojet/CRN framework provides a practical pathway to:

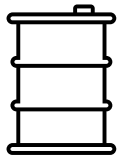
- Interpret emissions **mechanisms** and understand dominant **drivers.**
- Quantify **sensitivities** to fuel properties and operating conditions.
- Evaluate **alternative fuels** using surrogate fuel formulations, based on **thermodynamic state evolution** rather than via external correlations.
- Generate **validated data** and model insight to support **extrapolation** to full-scale turbofan engines.

[#] (Sharma et al. 2026), [#] (Lieder et al. 2026), [#] (Goeing et al. 2023)

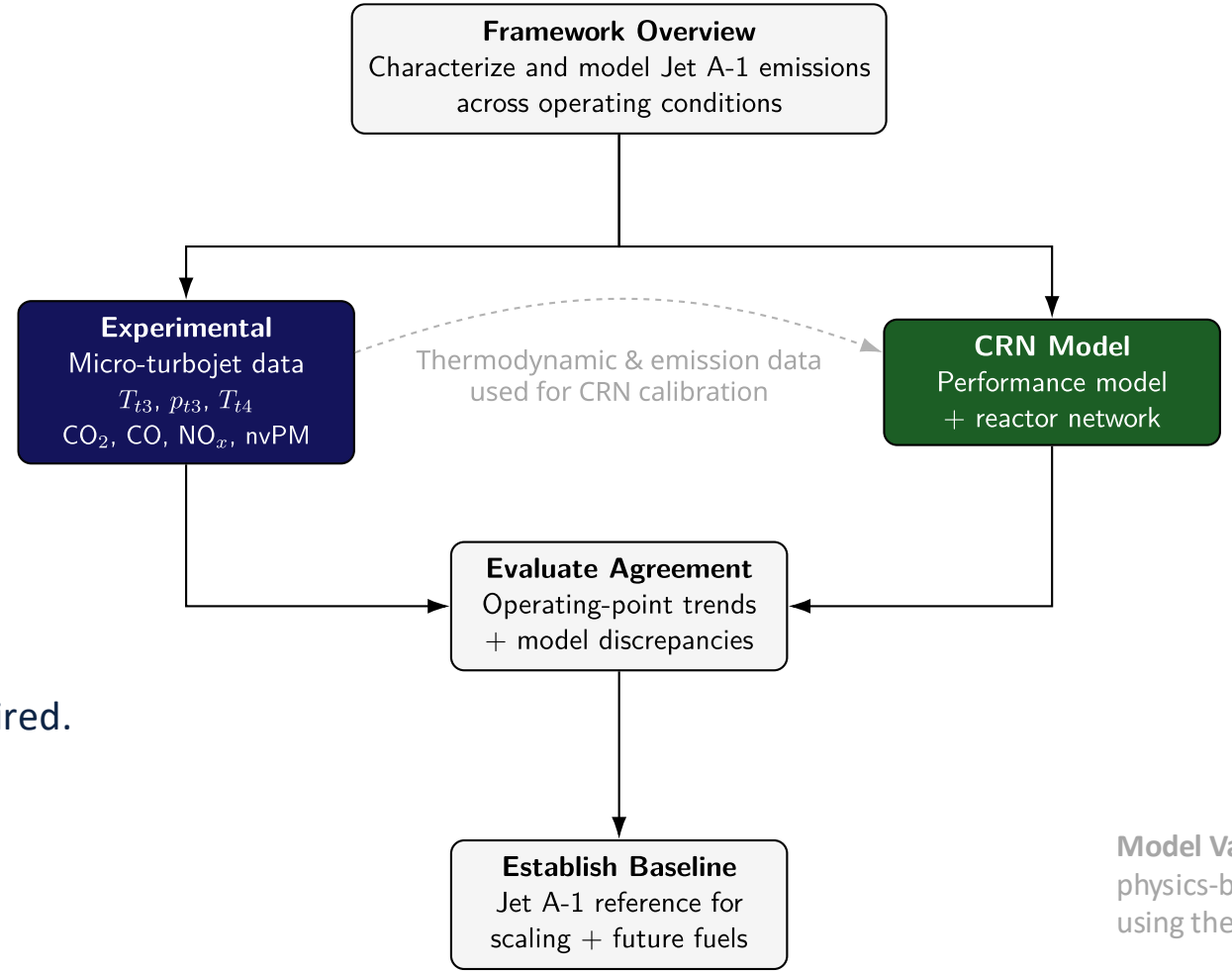
Research Objectives

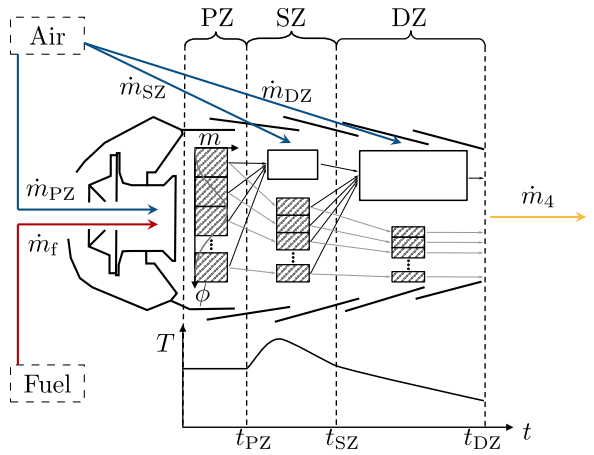
To what extent can the coupled framework characterize the Jet A-1 operating-point gaseous emission indices, and the nvPM number size distribution, and where do the main discrepancies between experiment and CRN model prediction remain?





➔ 50-100 L of fuel required.





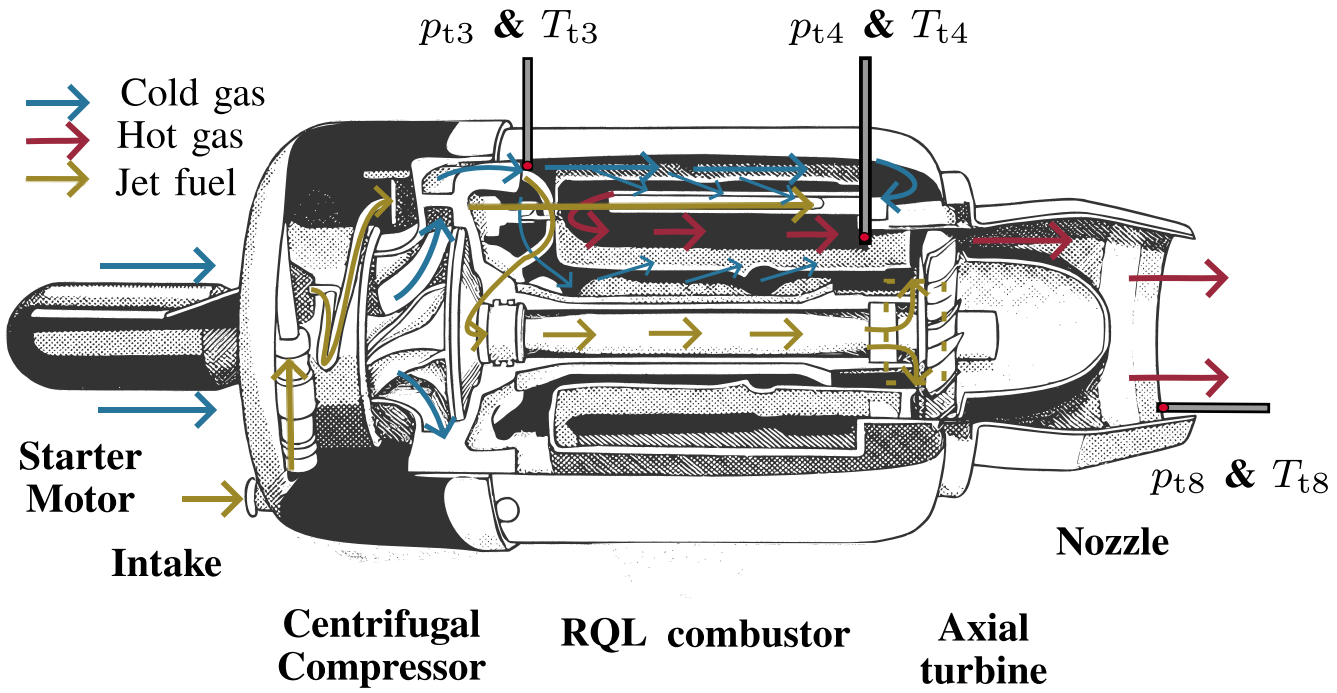
PFR
PSR
Air at T_{t3}
Fuel
Mixed gas
Connections

PZ →Primary Zone
SZ →Secondary Zone
DZ →Dilution Zone
$\dot{m}_{PZ,SZ,DZ}$ →Air mass flows
$\dot{m}_4$ →Outlet mass flow
$\dot{m}_f$ →Fuel mass flow

Model Validation: Evaluate the predictive accuracy of a physics-based Chemical Reactor Network (CRN) model using the generated micro-turbojet emissions data.

METHODOLOGY: THE MICRO-TURBOJET ENGINE

JetCat P200-SX Micro-Turbojet:



- The engine utilizes **open-loop** lubrication system.

Engine Specifications:

Max RPM	112,000
Max Thrust (N)	230
Pressure Ratio	4
Mass Flow (kg/s)	0.45
Fuel Flow @ Max RPM (kg/s)	0.0096
Nozzle Exit Area (m^2)	0.0028

METHODOLOGY: EXPERIMENTAL SETUP

Performance Measurements

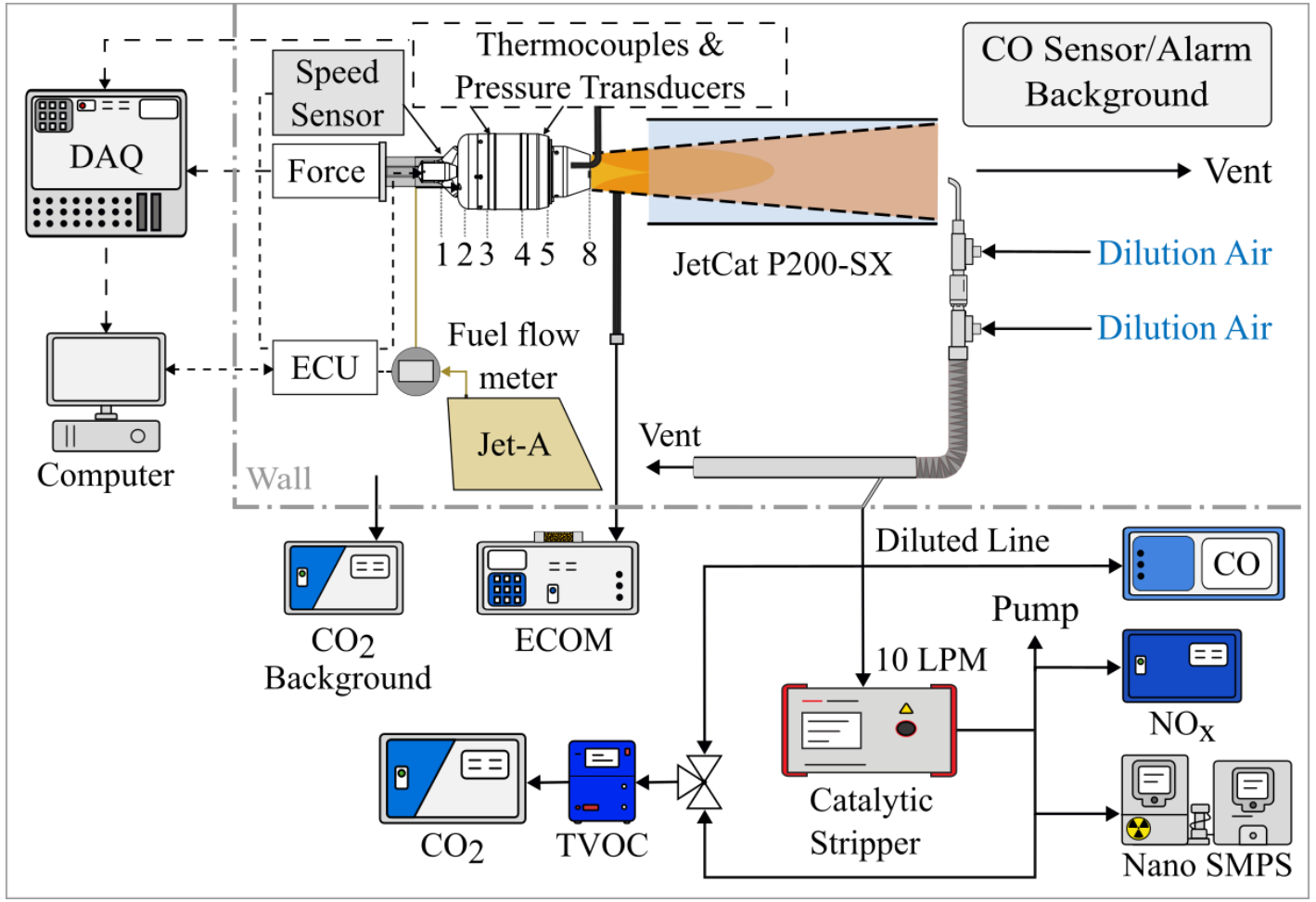
- Thermocouples, pressure transducers, a force transducer, speed sensor, and fuel flow metering were used to track engine operating condition.

Undiluted Line (Jet Core)

- Prandtl probe at nozzle exit.
- ECOM measures raw CO_2, CO, and NOx.

Diluted Line

- Prandtl probe with 2-stage ejector dilutor.
- **Gas-Phase Analysis**
  - Dedicated NOx, CO, and CO_2 analyzers on the diluted line.
- **Particle Phase**
  - Measures via SMPS (nano-DMA) downstream of a **Catalytic Stripper**. The stripper removes semi-volatile organic compounds (SVOCs) allowing us to isolate non-volatile PM (soot).



METHODOLOGY: PHYSICS BASED MODELING

Reduced-order physics-based framework

- Captures fuel-dependent and operating-point-dependent emissions sensitivities across multiple fuels and engine operating conditions.
- Altitude effects are represented physically

Aero model provides combustor inlet state:

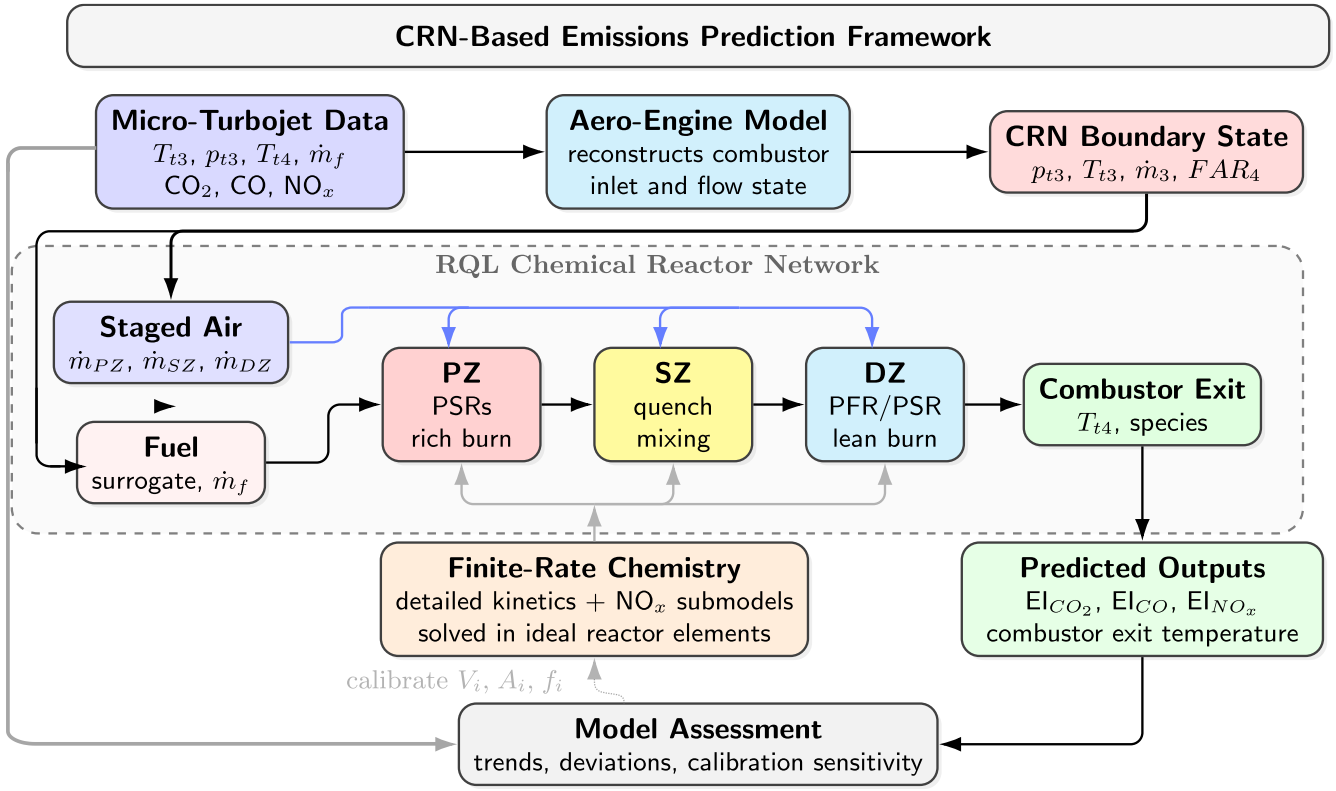
- p_{t3}, T_{t3}, $\dot{m}_3$, FAR_4

RQL combustor represented using ideal reactors:

- PSR → quench/mixing → PFR/PSR
- Finite-rate chemistry predicts CO_2, CO, NO_x, nvPM

- Calibrated and validated using experimental engine data
- Enables fuel-and operating-point-sensitive emissions prediction

Feasible for large emissions assessments



$$A_{PZ} = \frac{\dot{m}_{PZ}}{\dot{m}_3},\ A_{SZ} = \frac{\dot{m}_{SZ}}{\dot{m}_3},\ A_{DZ} = \frac{\dot{m}_{DZ}}{\dot{m}_3} \qquad \tau_i \sim \frac{V_i}{\dot{m}_i} \qquad \phi_{PZ} = \frac{\dot{m}_f}{A_{PZ}\,\dot{m}_3\,FAR_{st}} \qquad A_{DZ} = 1 - A_{PZ} - A_{SZ}$$

METHODOLOGY: CALCULATION OF EMISSION INDICES

A. EXPERIMENTAL DATA: CARBON-BALANCE METHOD



Inputs (measured at diluted line)

$x^{DL}_{CO_2}, x^{DL}_{CO}, x^{DL}_{NO_x}$ (dry mole fractions)

$\frac{dN_{nv,corr}}{d\log_{10} d_p}$ (corrected nvPM number size distribution)

$x^{bg}_{CO_2}$ (background CO_2)

1 Gaseous species EI (generic formula)

$$EI_i\ [\text{g/kg}_{\text{fuel}}] = \frac{1000 M_i\, \eta_{comb} Y_C}{M_C} \frac{x_i^{DL}}{(x^{DL}_{CO_2} - x^{bg}_{CO_2}) + x^{DL}_{CO}}$$

with $i = CO_2, CO, NO_x$

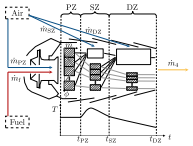
2 nvPM number EI (size-resolved)

$$\frac{dEI_{nvPM,N}}{d\log_{10} d_p}\ [\#/\text{kg}_{\text{fuel}}] = \frac{10^{12}\ \frac{dN_{nv,corr}}{d\log_{10} d_p}\ \eta_{comb} Y_C R T_{ref}}{\left[(x^{DL}_{CO_2} - x^{bg}_{CO_2}) + x^{DL}_{CO}\right] M_C p_{ref}}$$

$dN_{nv,corr}/d\log_{10} d_p$: nvPM number size distribution corrected for dilution, thermophoretic and size-dependent losses [#/cm^3]

B. CRN MODEL: CRN EI CALCULATION METHOD

Inputs (from aero-engine model & CRN)



$\dot{m}_{i,out}, Y_{i,out}$ (gas mass fractions),

$\frac{d\dot{N}_{soot}}{d\log_{10} d_p}$ (soot number size distribution)

$\dot{m}_f$ (fuel mass flow rate)

1 Gaseous species EI

$$EI_i\ [\text{g/kg}_{\text{fuel}}] = 1000 \frac{\dot{m}_{i,out}}{\dot{m}_f}$$

$$\dot{m}_{i,out} = Y_{i,out} \dot{m}_{out}$$
with $i = CO_2, CO, NO_x$

2 nvPM number EI (size-resolved)

$$\frac{dEI_{nvPM,N}}{d\log_{10} d_p}\ [\#/\text{kg}_{\text{fuel}}] = \frac{1}{\dot{m}_f} \frac{d\dot{N}_{soot}}{d\log_{10} d_p}$$



CRN soot model (sectional approach):

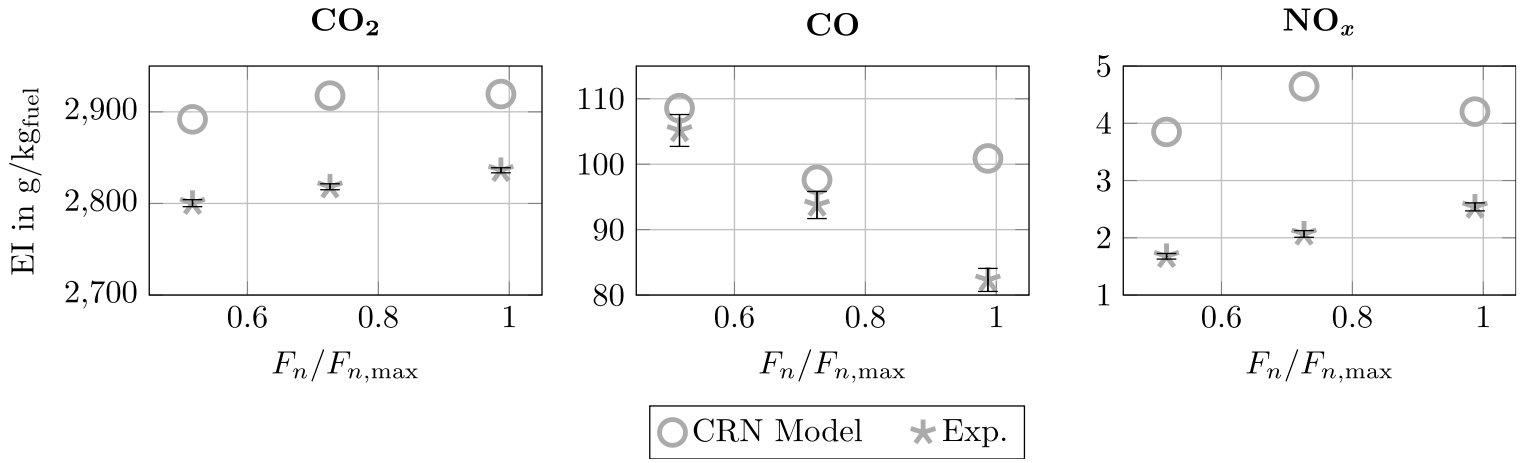
Finite-rate chemistry coupled with a discrete sectional soot model (21 size bins; $d_p \approx 2.02$ nm to 202.12 nm) predicts soot number and mass across particle sizes.

RESULTS: GASEOUS EI

CRN model captures **overall trends** across the operating range.

CO_2 and NO_x increasing with thrust, while **CO decreases** with increasing thrust.

Model overpredicts **NO_x**, likely due to temperature inhomogeneities that are not represented in the idealized reactor network.



ABSOLUTE PERCENTAGE DEVIATION OF CRN MODEL AGAINST EXPERIMENT

Species	Metric	Normalized Net Thrust ($F_n/F_{n,max}$)		
		50%	70%	100%
CO_2	Abs. Deviation (%)	2.9	3.2	3.4
CO	Abs. Deviation (%)	3.6	5.0	22.9
NO_x	Abs. Deviation (%)	118	127	78

Abs. Deviation (%) = $|CRN - Exp.| \,/\, Exp. \times 100$.

The CRN model reproduces CO2 and CO within $\pm 5\%$, except for CO at 100% thrust ($\approx 23\%$). NOx is systematically high (up to $\sim 130\%$) but retains the correct positive thrust sensitivity. The largest deviations are consistent with the reduced PSR/PFR formulation, where local temperature inhomogeneity, mixing non-uniformity, and residence-time variation are compressed into idealized reactor elements.

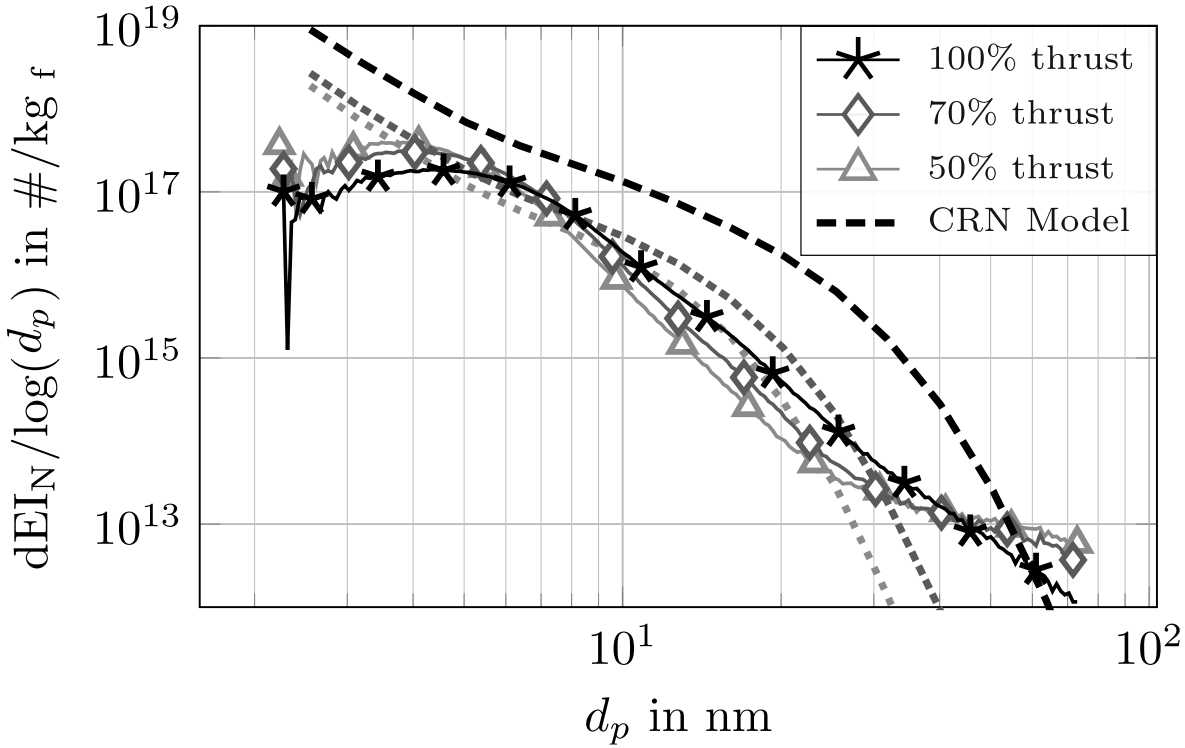
RESULTS: NON-VOLATILE PM EI

Similar shape & trends:

- CRN model reproduces the overall nvPM EI particle size distributions from experiment.
- Distributions shift right (larger mobility diameters) for increasing thrust in both experiment and model.
- Predicted d_p remain within the same order of magnitude as the measurements across much of the distribution.

Useful for comparative studies:

- Establishes a consistent Jet A-1 reference case for future fuel comparisons.
- Enables relative assessment of operating-point and fuel-dependent changes.
- Captures whether particle number distributions shift in magnitude, shape, or modal diameter.
- Provides a tractable framework for screening fuel sensitivities before higher-fidelity modeling or expanded testing.



At the smallest sizes, uncertainty in the experimental loss correction contributes substantially to the discrepancies between model and experiment. At larger sizes, the differences are likely influenced by model assumptions, including the simplified 1-D PSR/PFR representation of the combustor, unresolved local mixing and residence-time variation, and the sectional soot model assumption of spherical particles with a single size-independent effective density.

Summary

- **Micro-turbojet facility** generated controlled Jet A-1 aero-thermodynamic performance, gaseous emissions, and nvPM number size distributions across a range of operating conditions.
- The **coupled aero-engine/CRN framework** captured key operating-point trends for CO_2, CO, NO_x and nvPM; strongest agreement was observed for CO_2 and CO.
- **Largest discrepancies** remain for NO_x emission indices and the magnitude of small-diameter nvPM, consistent with current reactor-network assumptions and loss-correction uncertainties.
- Results establish a **Jet A-1 baseline dataset** for assessing the coupled framework and identifying priority areas for model improvement.

Future Work

1. **Expand experimental database**
   - Conduct back-to-back fuel-comparison studies with SAF / FT-SPK blends and other alternative fuels, with more operating points.
2. **Refine model and reduce uncertainties**
   - Improve CRN treatment of temperature inhomogeneity, mixing, residence time, and soot inception.
   - Develop effective density as a function of mobility diameter within the CRN framework.
   - Update experimental setup, and exhaust conditioning
   - Incorporate additional diagnostics and hardware (e.g. FTIR, AMS, TEM), to better characterize the experimental data.
3. **Scaling**
   - Use the assessed micro-turbojet + CRN framework to support scaling toward full-size aero-engine emissions assessment.
   - Quantify sensitivities relevant to contrail-formation potential and climate-impact models, including nvPM number, size, and composition.

ACKNOWLEDGMENTS

- UBC Clean Energy Research Center (CERC)
- The NSERC Discovery Grant Program
- The Deutsche Forschungsgemeinschaft (DFG, German Research Foundation) under Germany's Excellence Strategy
- Sustainable and Energy Efficient Aviation and the TUBS Postdoc International Mobility Grants.

Supervisor: Dr. Steven Rogak

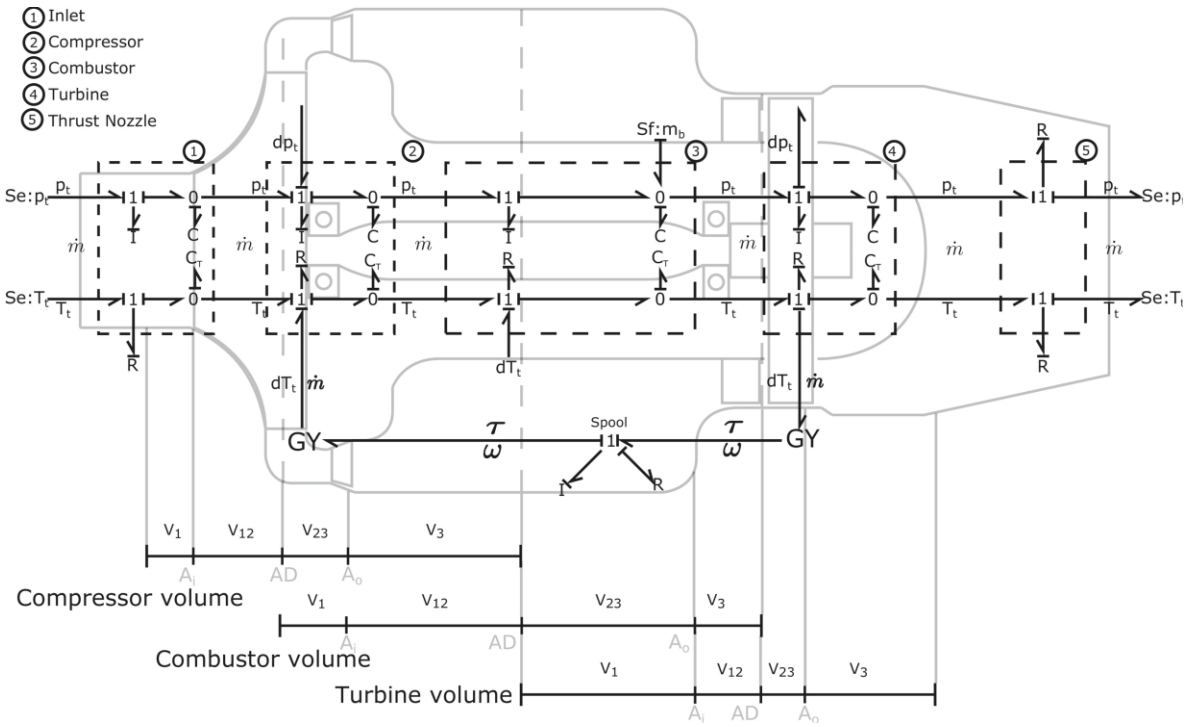
Individuals: Dr. Patrick Kirchen, Dr. Jan Göing, Daniel Leider (PhD Student TUBS), and everyone at CERC for letting us operate a jet engine!

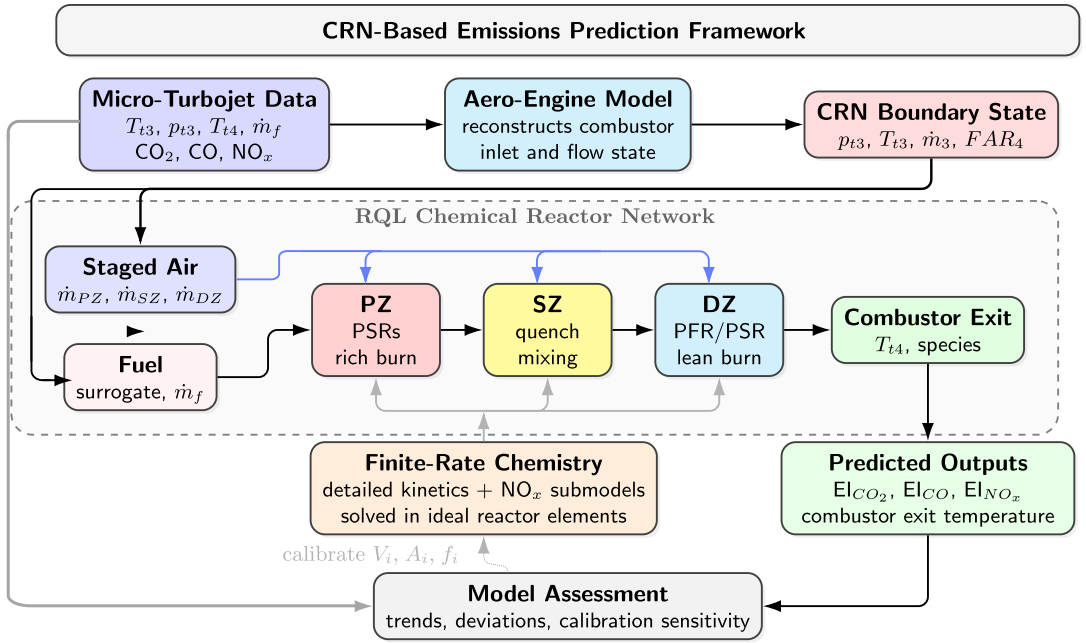
THANK YOU, QUESTIONS?

APPENDIX

APPENDIX: PHYSICS BASED MODELING

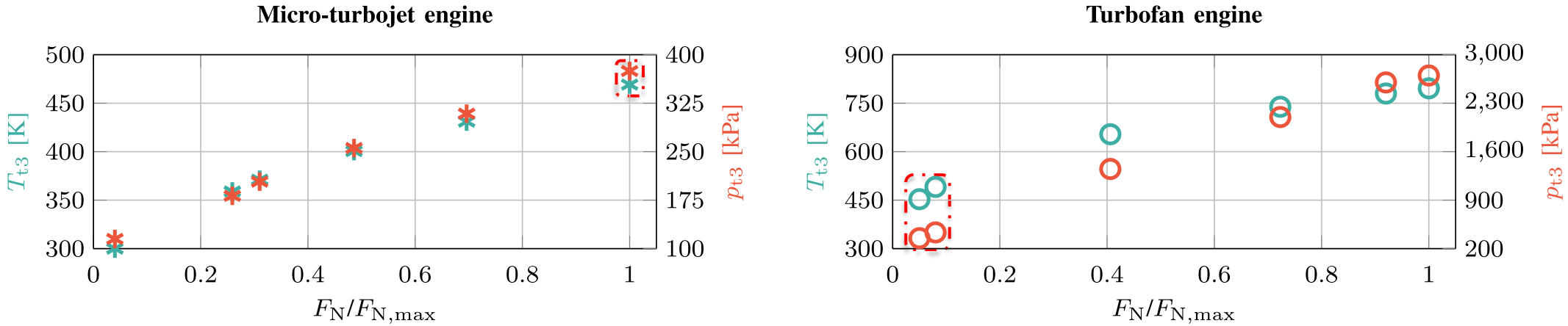
Aero Engine Model (Goeing et al. 2020)





$$A_{PZ} = \frac{\dot{m}_{PZ}}{\dot{m}_3},\ A_{SZ} = \frac{\dot{m}_{SZ}}{\dot{m}_3},\ A_{DZ} = \frac{\dot{m}_{DZ}}{\dot{m}_3} \qquad \tau_i \sim \frac{V_i}{\dot{m}_i} \qquad \phi_{PZ} = \frac{\dot{m}_f}{A_{PZ}\,\dot{m}_3\,FAR_{st}} \qquad A_{DZ} = 1 - A_{PZ} - A_{SZ}$$

APPENDIX: MICRO-TURBOJET VS TURBOFAN



- Direct operating-point comparison between the JetCat and a full-scale V2500 turbofan is constrained by scale. However, a specific thermodynamic intersection exists:

Engine	OPR	T_{t3}	P_{t3}
Micro-Turbojet @ max OP	≈ 4	**469 K**	**374 kPa**
Full size Turbofan @ idle OP [14]	$\approx 3.5 - 4$	453 K	351 kPa

"Micro-Max / Macro-Idle": This thermodynamic intersection establishes a physically valid anchor point for the performance. **Matching combustor inlet conditions creates a similar environment for combustion**, enabling valid comparisons. It allows us to approximate commercial turbofan idle conditions in a laboratory setting.

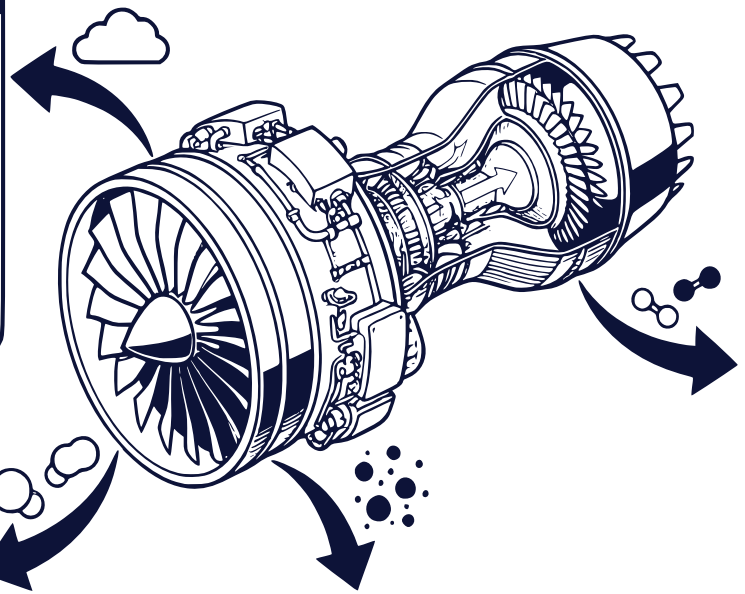
[14] (Göing 2023)

Appendix: Aero Engine Emissions

Products of Complete Combustion (Greenhouse Gases):



Carbon dioxide (CO_2) & Water vapor (H_2O): are the primary natural byproducts of hydrocarbon combustion. Both are GHGs. H_2O plays a critical role in contrail formation at high altitudes.



Products of Incomplete Combustion (Lower Power):



Carbon Monoxide (CO) & Unburned Hydrocarbons (UHC): Greatest concentrations predominantly generated during low-power operation (e.g., idle), where combustion efficiency is lowest.

High-Temperature Byproducts (Higher Power):



Oxides of Nitrogen (NOx), (NO & NO_2): formed via the thermal reaction of atmospheric **N_2** and **O_2** at extremely high temps (**>1800 K, Thermal (Zel'dovich) mechanism**), during high-power settings like take-off and climb.

Particulate Matter (PM) & Impurities:



Non-volatile PM (nvPM): Ultrafine black carbon (soot) particles formed in fuel-rich regions of the combustor.
Volatile PM (vPM): Formed when gaseous precursors (like volatile organic compounds) cool and condense after exiting the engine.

Aero engine emissions (depend on flight phase, & fuel composition)

Appendix: Aero Emissions Regulations

How do international bodies regulate these emissions and the fuels that produce them?

The Regulatory Landscape:

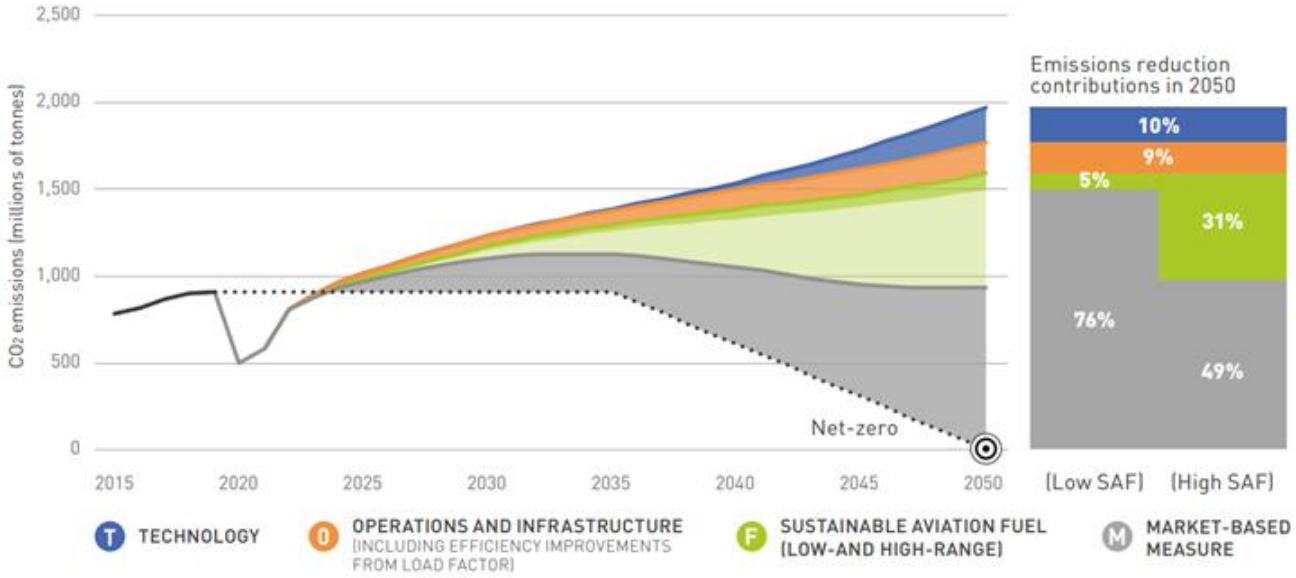
- **Engine Certification (ICAO Annex 16):** Standards for NO_x, nvPM, CO, and UHC are set by the CAEP (Committee on Aviation Environmental Protection) and enforced by national authorities like the Federal Aviation Administration (FAA) or Transport Canada Civil Aviation (TCCA) [8].

- **Fuel Certification (ASTM International): *ASTM D1655:** The standard for conventional Jet A/A-1 [36].
    - **ASTM D7566:** The critical specification for Sustainable Aviation Fuels (SAF), certifying them as "drop-in" ready [35].

ASTM D1655 – 25a: Detailed requirements of certified aviation fuels [36].

APPENDIX: SUSTAINABLE AVIATION FUELS

The Near-Term Solution: Sustainable aviation fuels (SAFs) are the primary viable pathway to decarbonize aviation, offering "drop-in" replacements sourced from biogenic or waste feedstocks [8, 35].



Industry targets to achieve "net zero" CO2 emission. [38].

- **Regulatory Hurdles:** Aviation operates under strict safety margins. Aircraft cannot legally fly on a new fuel unless it has rigorous ASTM International certification (e.g., ASTM D7566) [35].
- **The Reality:** Many promising, highly sustainable SAF pathways are not yet certified for commercial use, meaning they cannot be deployed [37].

[8] (ICAO Annex 16), [35] (ASTM D7566-25a), [37] (ICAO 2017), [38] (ATAG 2021)

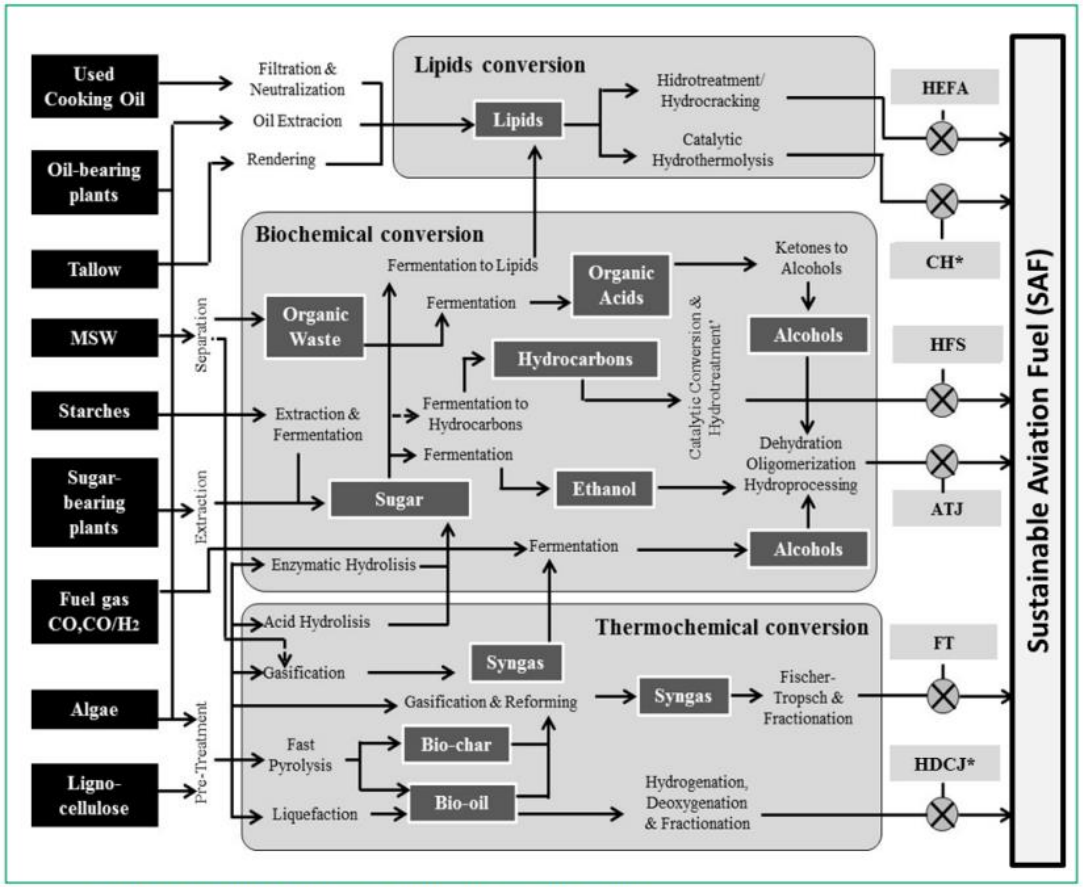
APPENDIX: SUSTAINABLE AVIATION FUELS

"Drop-In": While marketed as "drop-in" replacements, SAFs are different from conventional Jet Fuel (Jet A-1) & one another. Their chemistry is dependent on the feedstock and processing pathway.

Key Chemical Differences:

- **SAFs differ in Aromatic Content**
- **SAFs differ in Hydrogen-to-Carbon (H/C) Ratio**
- **SAFs differ in Volatility**

The Impact: The chemical composition dictate the emissions. Because SAFs are all different, we cannot assume any two SAFs will behave the same way in an engine.



*CH: Catalytic hydrothermolysis – not yet approved
HDCJ: Hydrotreated depolymerized cellulosic to jet – not yet approved

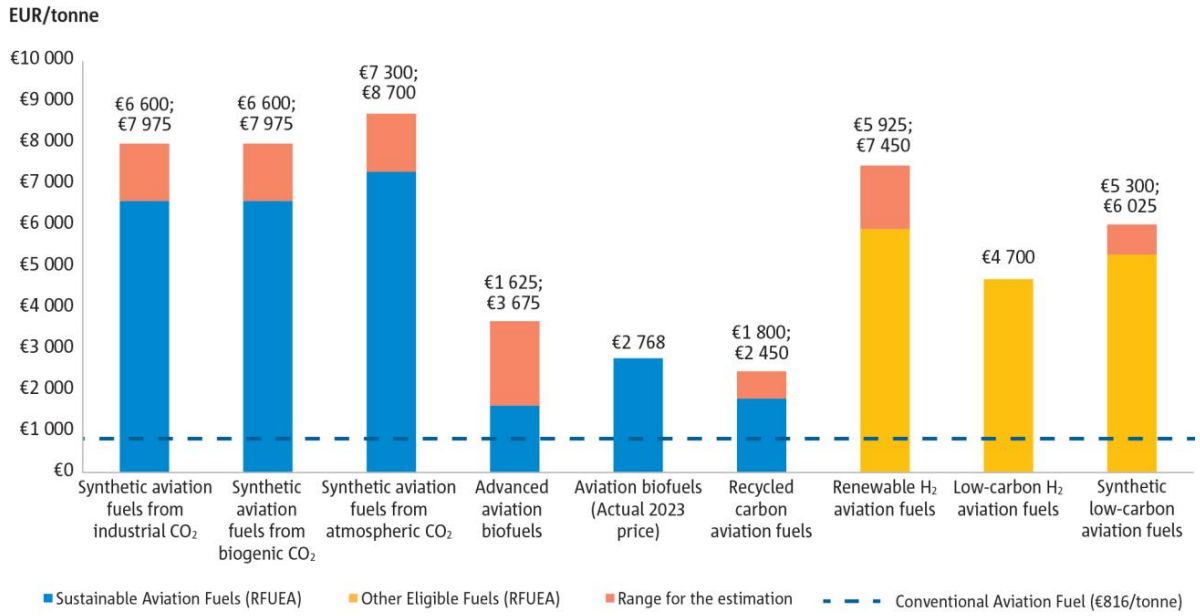
ICAO – Sustainable Aviation Fuels Guide: Different processing routes can be adopted to convert feedstocks into a liquid drop-in SAF. Several combinations of feedstocks and conversion processes have been proposed. Only some of them have been certified to be used as a drop-in fuel. [37].

[8] (ICAO Annex 16), [35] (ASTM D7566-25a), [37] (ICAO 2017)

APPENDIX: SUSTAINABLE AVIATION FUELS

To characterize the emissions from fuels we need extensive data on how they burn. Gathering this emissions data using full-size commercial turbofans is a major bottleneck:

Regulatory constraints: Strict safety protocols make testing experimental, uncertified fuels on large commercial hardware incredibly restrictive.



Estimated prices and production costs in 2023 for ReFuelEU Aviation eligible fuels [39].

Fuel Intensive: Requires tens of thousands of liters of experimental fuel, which is nearly impossible to produce for early-stage novel SAFs.

Prohibitively Expensive: These novel SAFs are orders of magnitude more expensive than Jet A-1 testing costs tens of thousands to hundreds of thousands of dollars per campaign for simply fuel costs, not accounting for the requisite hardware to burn and characterize these fuels.

[39] (EASA 2022)

APPENDIX: AERO ENGINE EMISSIONS

Aero engine emissions (depend on flight phase, fuel composition, engine architecture):

- **Products of Complete Combustion (Greenhouse Gases):**
  - Carbon dioxide (CO_2) and water vapor (H_2O) are the primary natural byproducts of hydrocarbon combustion.
  - Both are greenhouse gases, with water vapor playing a critical role in the formation of contrails at high altitudes.
- **Products of Incomplete Combustion (Low Power):**
  - Carbon Monoxide (CO) and Unburned Hydrocarbons (UHC).
  - These pollutants are predominantly generated during low-power engine settings, where combustion efficiency is lowest.
- **High-Temperature Byproducts (High Power):**
  - Oxides of Nitrogen (NOx), comprising NO and NO_2.
  - NOx is formed via the thermal reaction of atmospheric nitrogen and oxygen at extremely high temperatures (above 1800 K, Thermal (Zel'dovich) mechanism), which typically occur during high-power settings like take-off and climb.
- **Particulate Matter (PM) and Impurities:**
  - Non-volatile PM (nvPM): Primarily consists of ultrafine black carbon (soot) particles formed in fuel-rich regions of the combustor.
  - Volatile PM (vPM): Formed when gaseous precursors (like volatile organic compounds) cool and condense after exiting the engine.

Appendix: Sustainable Aviation Fuels

"Drop-In": While marketed as "drop-in" replacements, SAFs are fundamentally different from one another. Depending on the feedstock and pathway, some are excellent, while others have incredibly poor combustion characteristics.

Key Chemical Differences:

Aromatic Content: Often much lower in SAFs. This reduces soot/nvPM formation but causes critical issues with engine seals shrinking and leaking.

Hydrogen-to-Carbon (H/C) Ratio: Typically higher, which fundamentally alters the flame temperature and structure in the combustor.

Lower Heating Value (LHV): Differences in energy density directly impact fuel flow rates and overall engine performance.

The Impact: These chemical shifts dictate the emissions. Because SAFs are all different, we cannot assume any two SAFs will behave the same way in an engine.

[37] (ICAO 2017)

APPENDIX: BACK-TO-BACK TESTING

Back-to-back testing: Consecutively testing different fuels under identical ambient and operating conditions to strictly isolate fuel-property effects from engine effects.

Why?

Isolates the Variables: Testing consecutively minimizes the impact of shifting environmental factors (temperature, humidity, atmospheric pressure) on combustion.

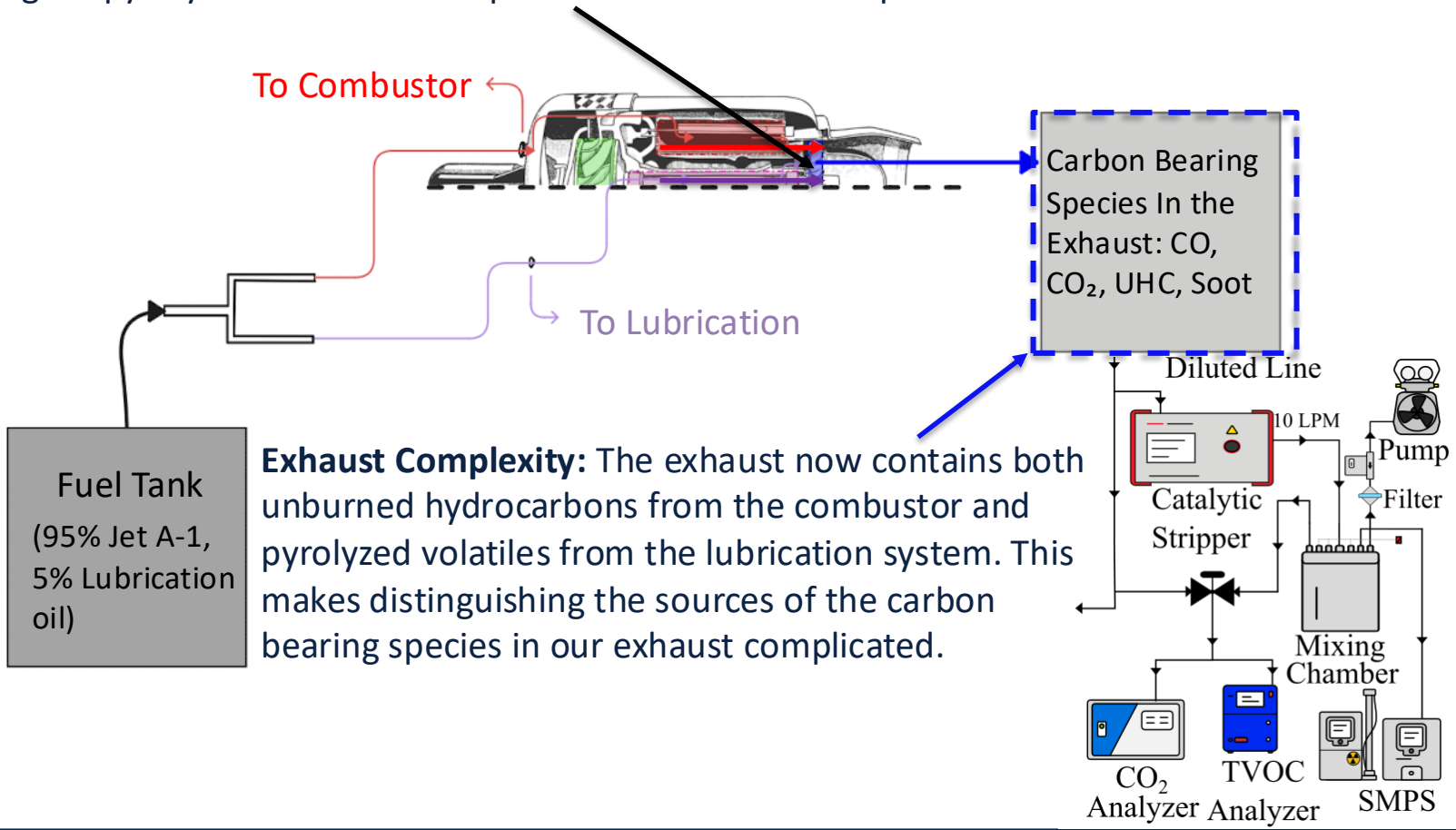
Proves "Fuel is the Factor": Ensures any observed shifts in emissions or performance are driven purely by fuel chemistry (e.g., aromatics, H/C ratio) and not by changes in engine operating conditions.

Captures Relative Sensitivities: Focuses on generating highly accurate differential data (e.g., EI_{CO} between SAF and Jet A-1) rather than just absolute magnitudes.

Enables Predictive Modeling: Provides the tightly controlled, physically grounded datasets required to accurately calibrate numerical scaling tools, such as Chemical Reactor Network (CRN) models.

APPENDIX: INDIRECT UHC MEASUREMENT

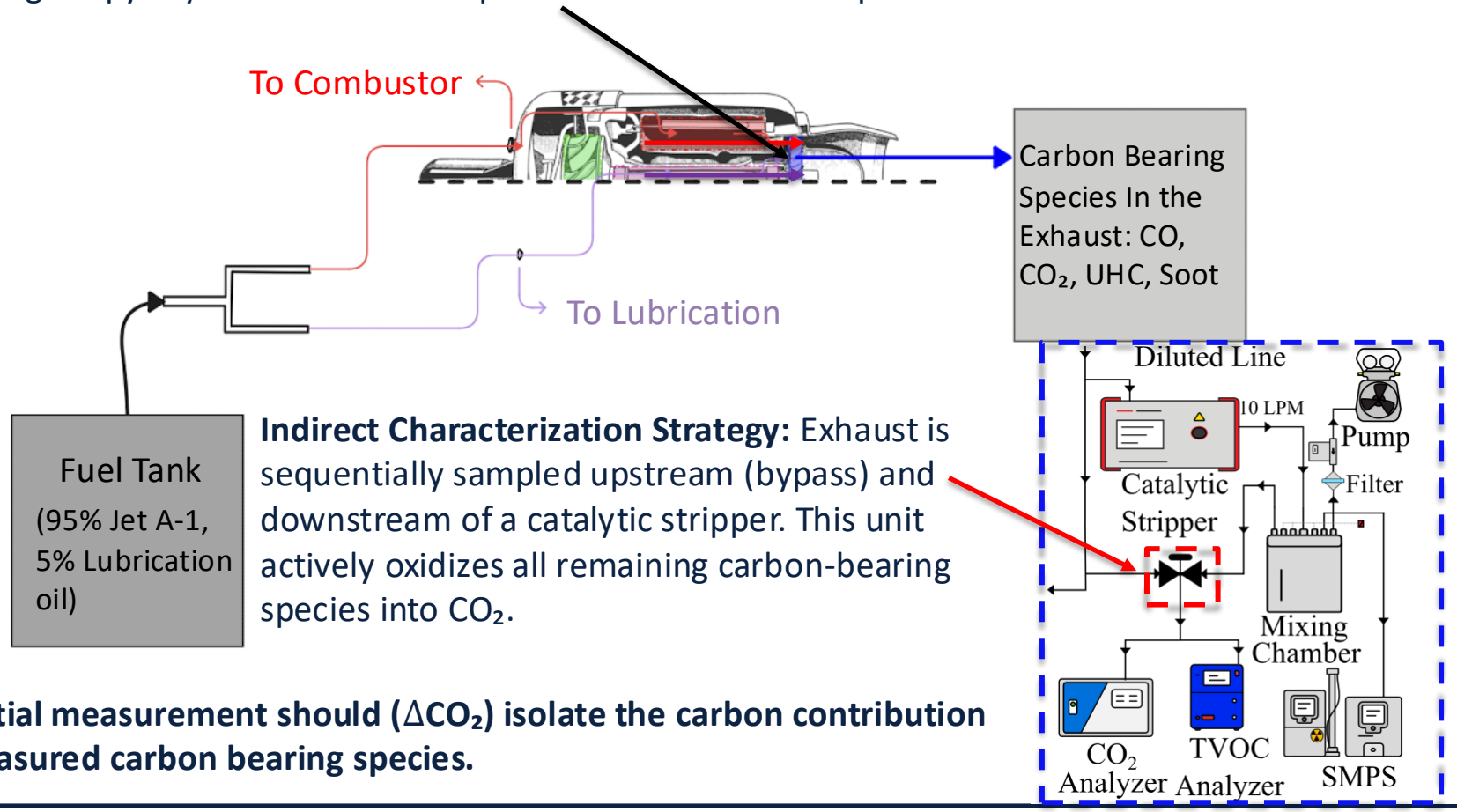
Open-loop: The system splits a 95% Jet A-1 / 5% lubrication oil blend between the combustor and the open-loop lubrication path. Lubricant is ejected directly into the gas path upstream of the axial turbine ($T_{t4} \approx 1200\ K - 1400\ K$). Within this high-temperature, oxygen-depleted environment, the lubrication oil likely undergoes pyrolytic thermal decomposition rather than complete oxidation.



Exhaust Complexity: The exhaust now contains both unburned hydrocarbons from the combustor and pyrolyzed volatiles from the lubrication system. This makes distinguishing the sources of the carbon bearing species in our exhaust complicated.

APPENDIX: INDIRECT UHC MEASUREMENT

Open-loop: The system splits a 95% Jet A-1 / 5% lubrication oil blend between the combustor and the open-loop lubrication path. Lubricant is ejected directly into the gas path upstream of the axial turbine ($T_{t4} \approx 1200\ K - 1400\ K$). Within this high-temperature, oxygen-depleted environment, the lubrication oil likely undergoes pyrolytic thermal decomposition rather than complete oxidation.



Indirect Characterization Strategy: Exhaust is sequentially sampled upstream (bypass) and downstream of a catalytic stripper. This unit actively oxidizes all remaining carbon-bearing species into CO_2.

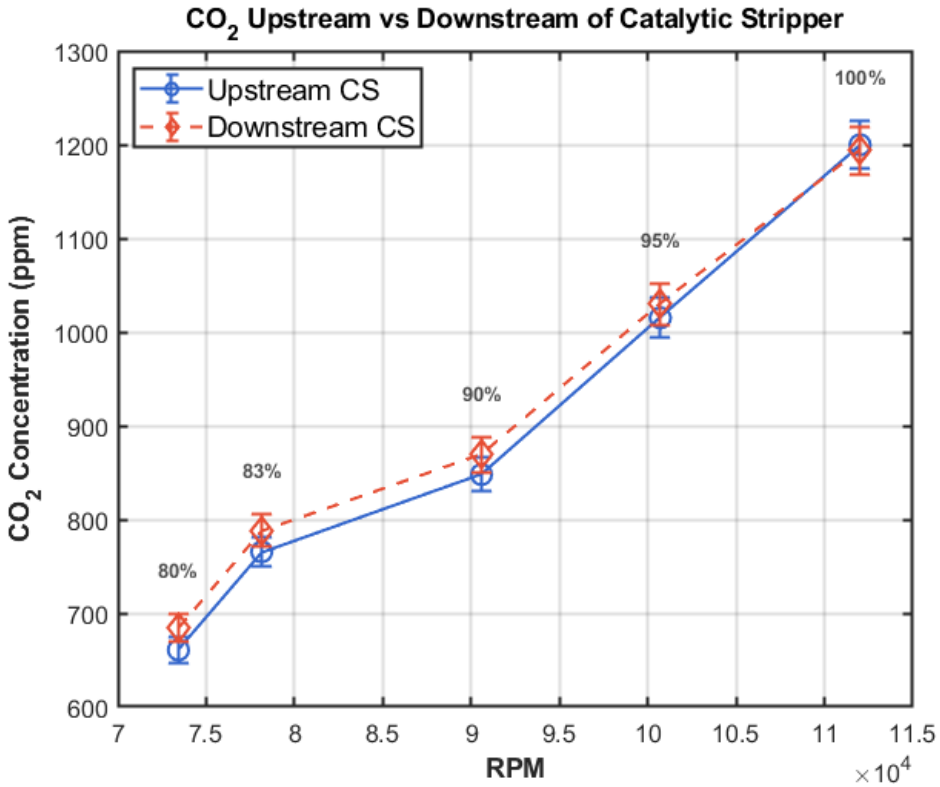
The differential measurement should (ΔCO_2) isolate the carbon contribution of the unmeasured carbon bearing species.

APPENDIX: INDIRECT UHC MEASUREMENT

- **The ΔCO_2 Signal:** Downstream CO_2 (red) is higher than upstream (blue) but decreasing with load, confirming the oxidation of unburned carbon species.

- **Load Dependency:** The measurable difference (ΔCO_2) is small and decreases as engine RPM increases.

It is interesting to note, that while the ΔCO_2 decreases with increasing RPM, the amount of lubrication oil flow rate also increases with the engine RPM.



Functional validation of the indirect UHC measurement system. Diluted exhaust gas is sequentially routed through a bypass line and a heated catalytic stripper using a 2-way switching valve. The stripper actively forces the complete oxidation of any surviving unburned fuel and pyrolyzed lubrication volatiles into measurable CO_2. This differential methodology isolates the total volatile carbon contribution without requiring complex chemical speciation.

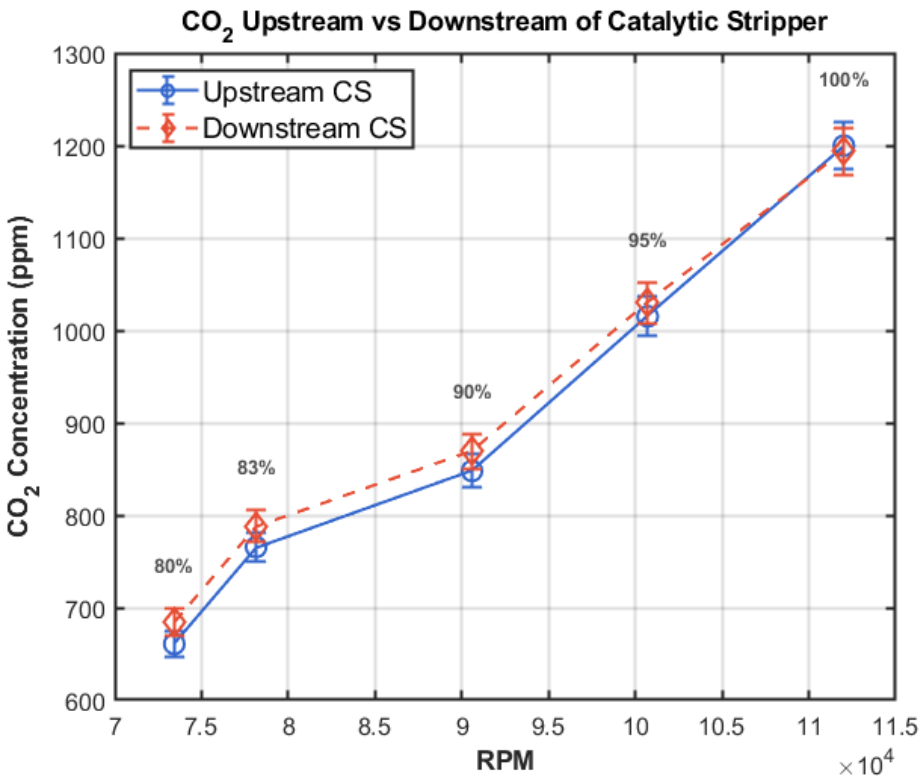
APPENDIX: INDIRECT UHC MEASUREMENT

System Split: Consistently, approx. 88% of the total flow reaches the primary combustor, while approx. 12% feeds the open-loop lubrication path.

At 100% thrust, the absolute mass of oil ejected into the exhaust is roughly four times higher than at idle (90 mL/min vs. 22 mL/min).

Thermal Oxidation: Despite this massive increase in injected oil, the ΔCO_2 gap shrinks to almost zero at max thrust.

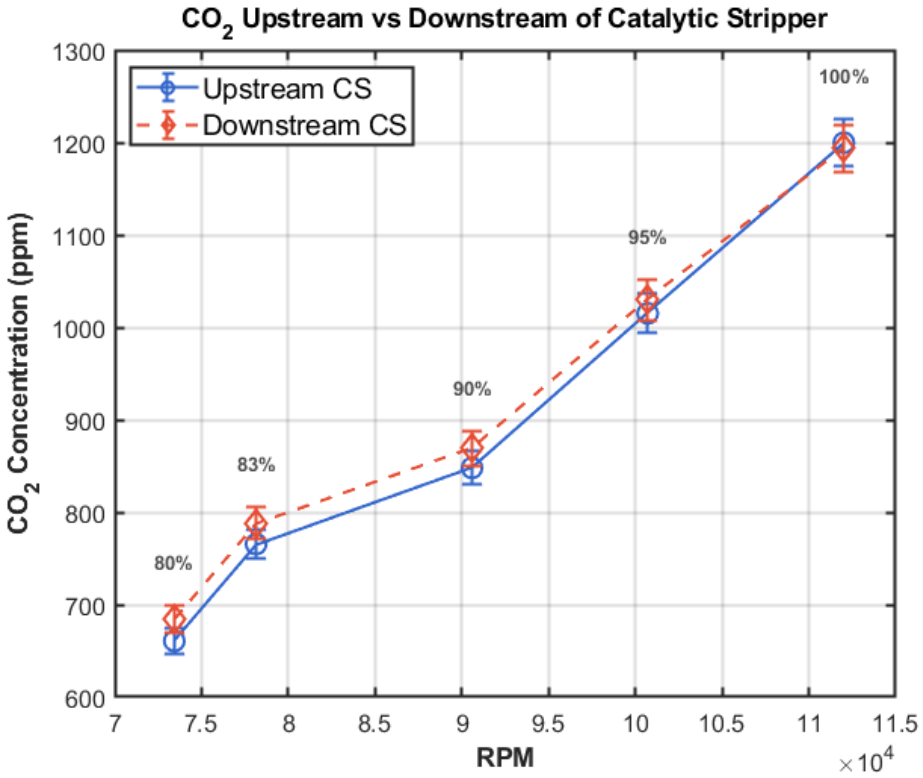
Physical Conclusion: At maximum load, extreme turbine inlet temperatures $T_{t4} \approx 1200\ K - 1400\ K$) likely cause the slipstream mixture to auto-ignite and fully oxidize before reaching the exhaust plane.



APPENDIX: INDIRECT UHC MEASUREMENT

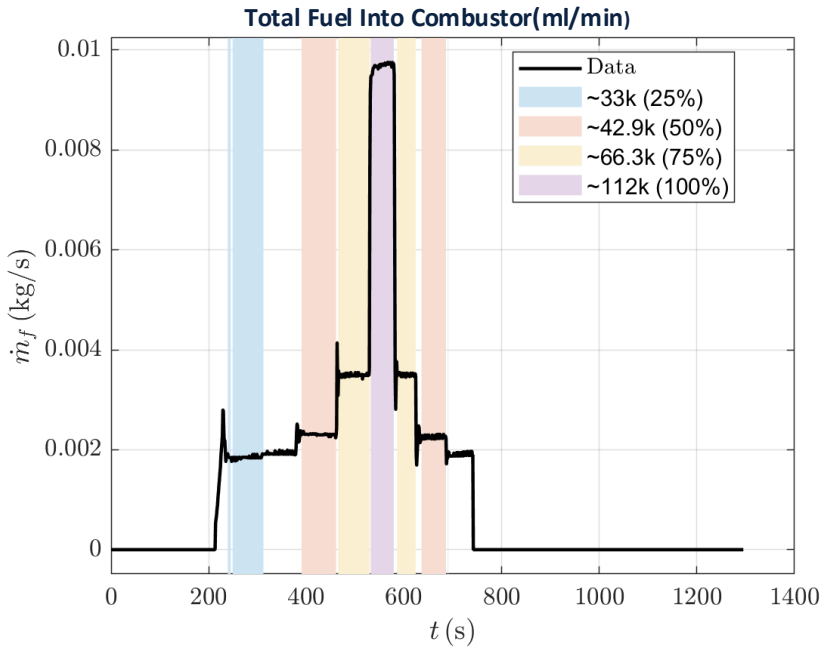
Low-Load Inefficiency is the Real Culprit: At lower RPMs (like 33k to 42k), temperatures are lower. The 22 mL/min of lubrication oil doesn't fully burn, and the primary combustor is operating inefficiently, dumping out primary UHCs. That's is likely why the ΔCO_2 gap is widest there. This is still speculation, we do not know for sure, but that's what I got out of this.

If 12% of the fuel was just slipping out of the engine unburned, the carbon balance would be off by 12%, and we had a max deviation of 6% at the lowest operating point.



Because the catalytic stripper proves the unmeasured carbon is negligible at high loads, we have proof that treating the exhaust as "fully oxidized" at the micro-max condition is a perfectly valid assumption.

Mass flow rate of Fuel to Combustor

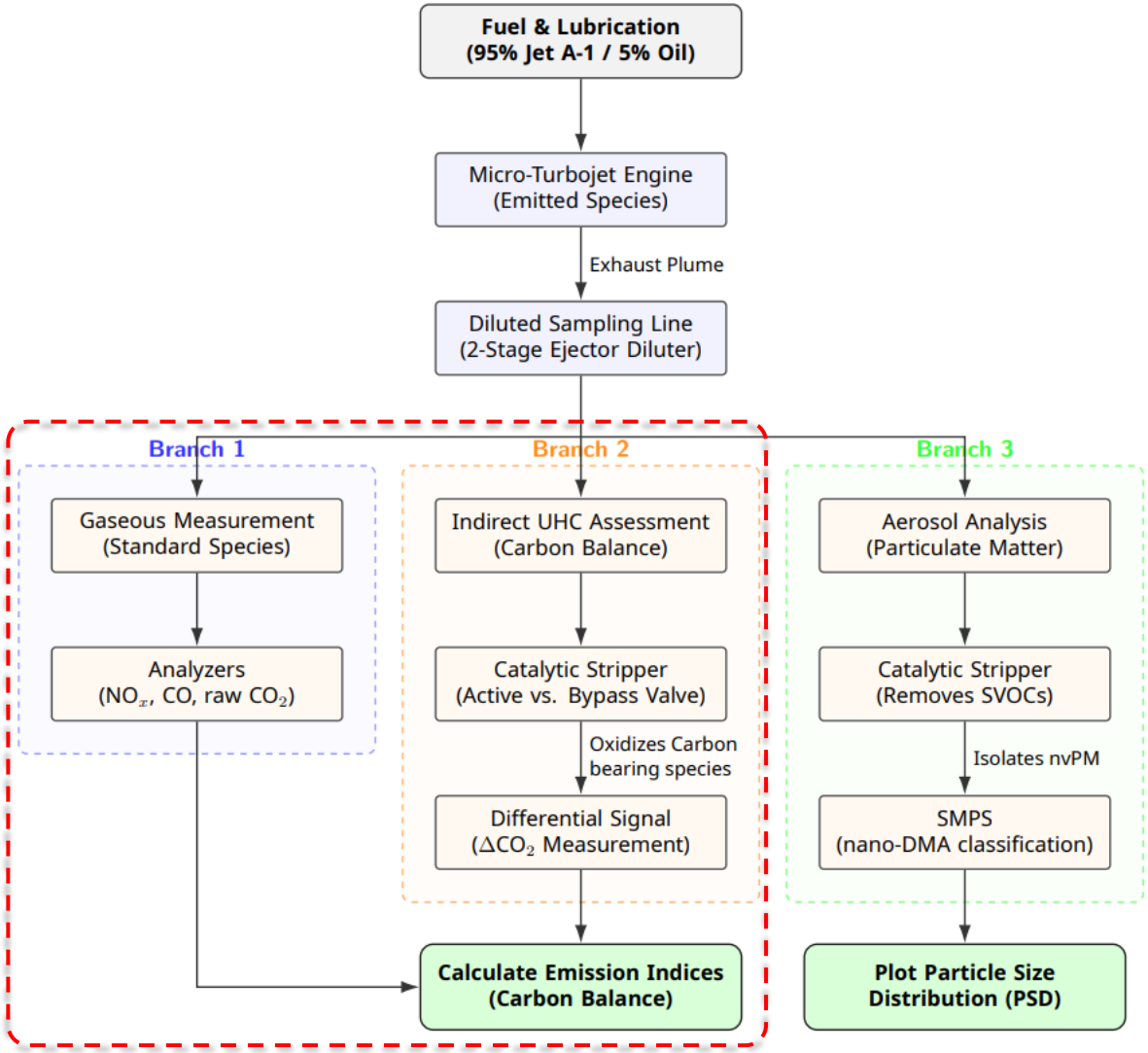


$$\dot{m}_f(t) = \rho_f \frac{\dot{V}_{f,\mathrm{comb}}(t)}{60000}, \quad \text{with } \rho_f \approx 0.80\text{–}0.82\,\mathrm{kg\,L^{-1}}$$

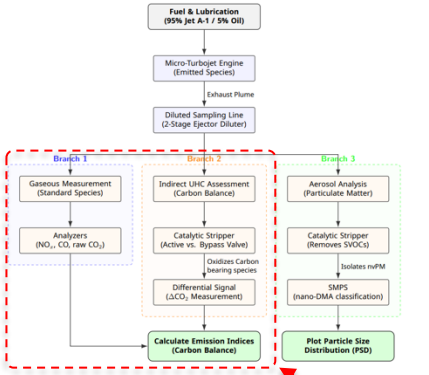
Thrust %	Total fuel [mL/min]	Combustor fuel [mL/min]
25	172	150
50	200	177
75	310	270
100	840	750

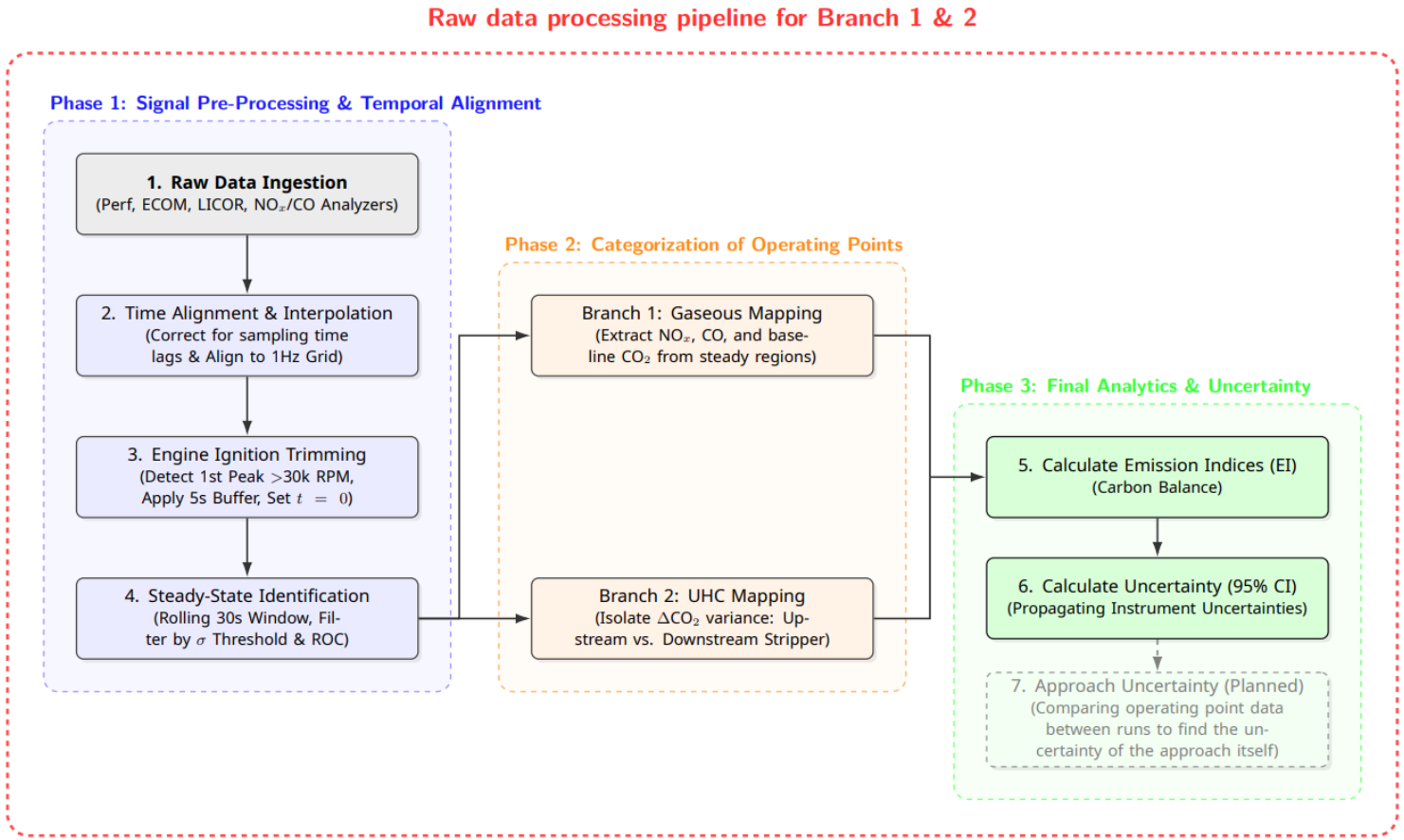
- Jan provided lookup data relating thrust command (%) to total fuel flow (mL/min)
- A fraction of this total flow actually reaches the combustor (≈ 87–89%)
- By comparing combustor flow to total flow across all setpoints, an average ratio of ~0.88 was established
- Converted volumetric flow (mL/min) to mass flow rate using fuel density (≈ 0.80–0.82 kg/L)
- This provided the fuel mass flow into the combustor, which is critical for equivalence ratio calculations

APPENDIX: DATA COLLECTION



APPENDIX: DATA PROCESSING





APPENDIX: PHYSICS SURROUNDING COMPARISON

Micro (JetCat)

- **Surface-to-volume:** High (significant wall quenching).
- Thus, **lower peak flame temperature**: Thermal NOx is exponentially dependent on temperature. We see **much lower EI NOx** for micro-turbojet which is consistent.
- Residence Time: ≈ 2 ms, this will limit CO oxidation.
- Max overall pressure ratio (OPR): 4

Macro (V2500)

- **Surface-to-Volume:** Low (Adiabatic core).
- Peak flame temperature **closer to adiabatic** flame temperature, much higher thermal NOx production, illustrated in **significantly higher in EI NOx** for turbofan.
- Residence Time: > 6 ms, assumed complete oxidation.
- Max overall pressure ratio (OPR): > 30

Direct operating-point-to-operating-point comparison across the thrust range is thermodynamically unjustified due to order-of-magnitude differences in scale. The sole meaningful correspondence exists at the operating envelope boundaries: "micro-max/macro-idle".

APPENDIX: PHYSICS SURROUNDING COMPARISON

	OPR	T_{t3}	P_{t3}
Micro-Turbojet @ max OP	≈ 4	469 K	374 kPa
Full size Turbofan @ idle OP [14]	≈ 3.5-4	453 K	351 kPa

$$k = A \cdot P^n \cdot exp(-\frac{E_a}{RT})$$

k (Reaction Rate Constant): The speed at which the chemical reaction (like soot nucleation) occurs.
A (Pre-exponential Factor): A constant that represents the frequency of molecular collisions.
P (Pressure): The local thermodynamic pressure (374 kPa).
n: An empirical constant showing how sensitive the reaction is to pressure changes.
E_a (Activation Energy): The minimum energy required to kickstart the chemical reaction.
R (Universal Gas Constant): A standard physical constant (8.314 J/(mol·K)).
T (Temperature): The local thermodynamic temperature (469 K).

At the “micro-max/macro-idle” intersection point, the combustor inlet air density, pressure, and temperature of the micro-turbojet and a full-scale turbofan are nearly identical. Combustion chemistry, specifically the formation of emissions that do not equilibrate in a gas turbine, depend entirely on how fast the chemical reactions are progressing (finite rate chemistry). For example, Nox, UHC, CO, Polycyclic Aromatic Hydrocarbons (soot pre-cursors) are driven local conditions and finite rate processes. This is also why a CRN model was used, finite rate chemistry is extremely important to modelling emissions accurately.

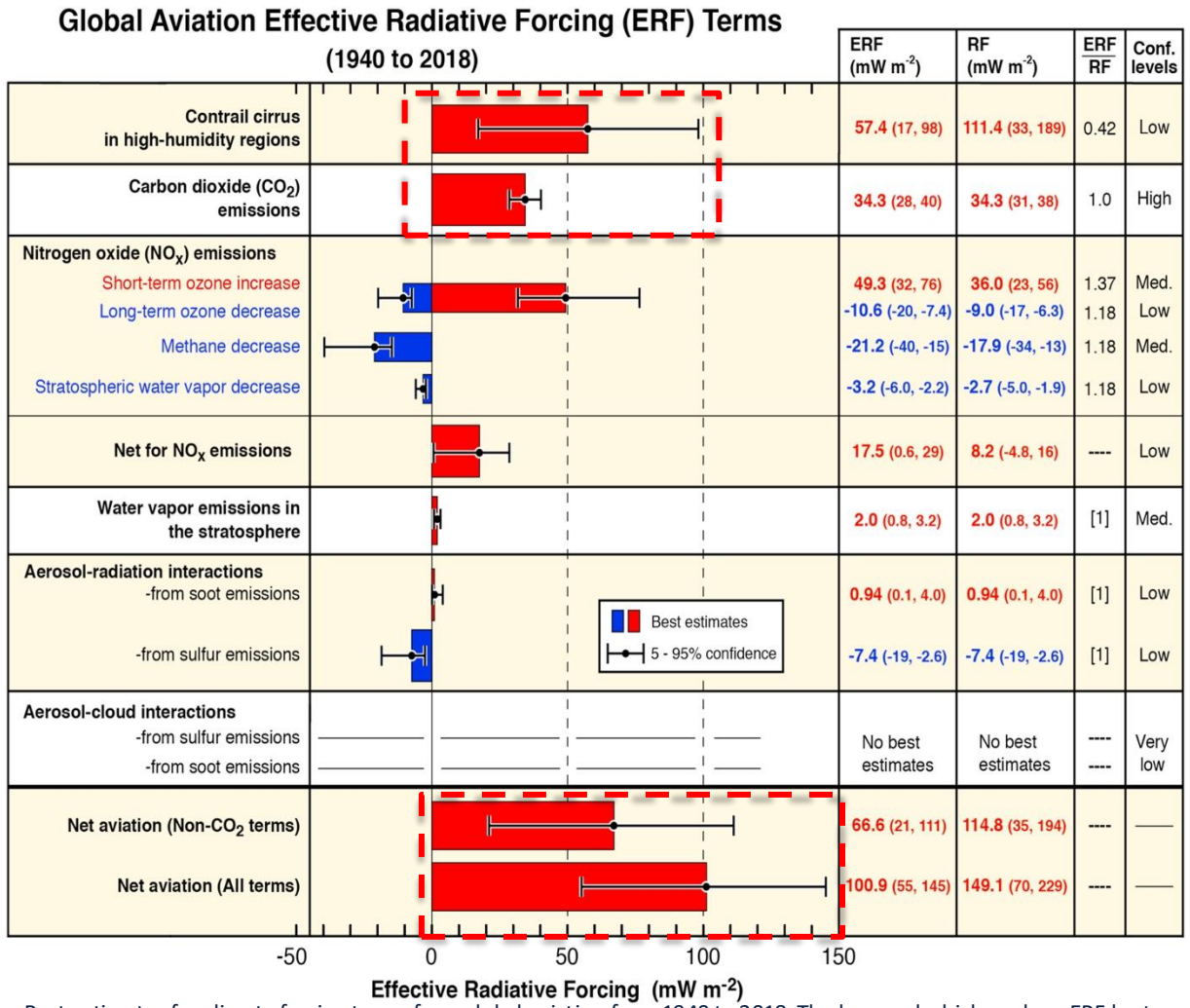
Non-CO_2 Forcing Mechanisms

Nitrogen Oxides (NO_x):

- Reactive nitrogen species drive near-field chemical transformations and interact with the atmosphere [5]

Particulate Matter & Contrails:

- Emitted soot can act as Ice Nucleating Particles (INPs), forming contrails & influencing the optical properties of clouds (albedo) which further amplifies radiative warming [3,7]



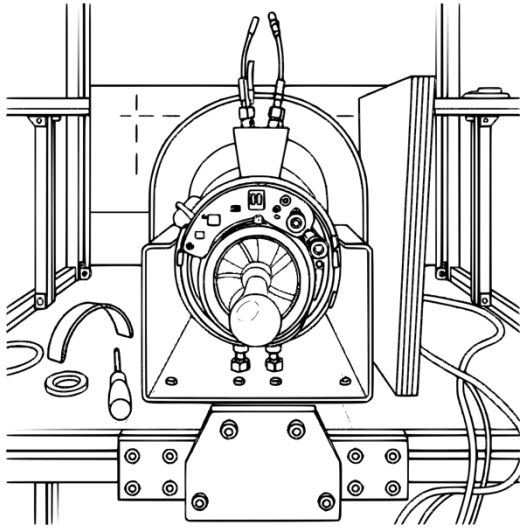
Best-estimates for climate forcing terms from global aviation from 1940 to 2018. The bars and whiskers show ERF best estimates and the 5–95% confidence intervals, respectively. Red bars indicate warming terms and blue bars indicate cooling terms [1].

[3] (Bickel et al. 2020), [5] (Bräuer et al. 2021) [6] (Tadic et al. 2021) [7] (Gettelman and Chen 2013)

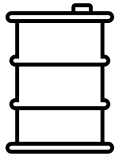
Research Objectives

Cost-effective, scaled-down laboratory facilities, such as an instrumented micro-turbojet test rig.

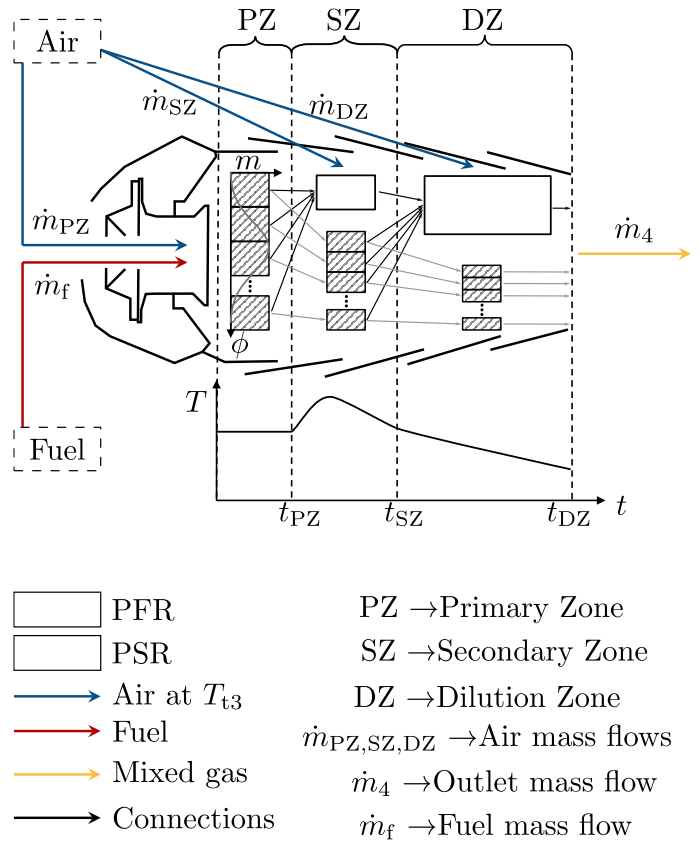
Benefit: Requires only modest fuel quantities
➔ Ideal for testing and characterizing various SAFs.



This capability facilitates the cost-effective acquisition of experimental data, enabling the exploration of operating-point and fuel-dependent sensitivities.



➔ 50-100 L of fuel required.

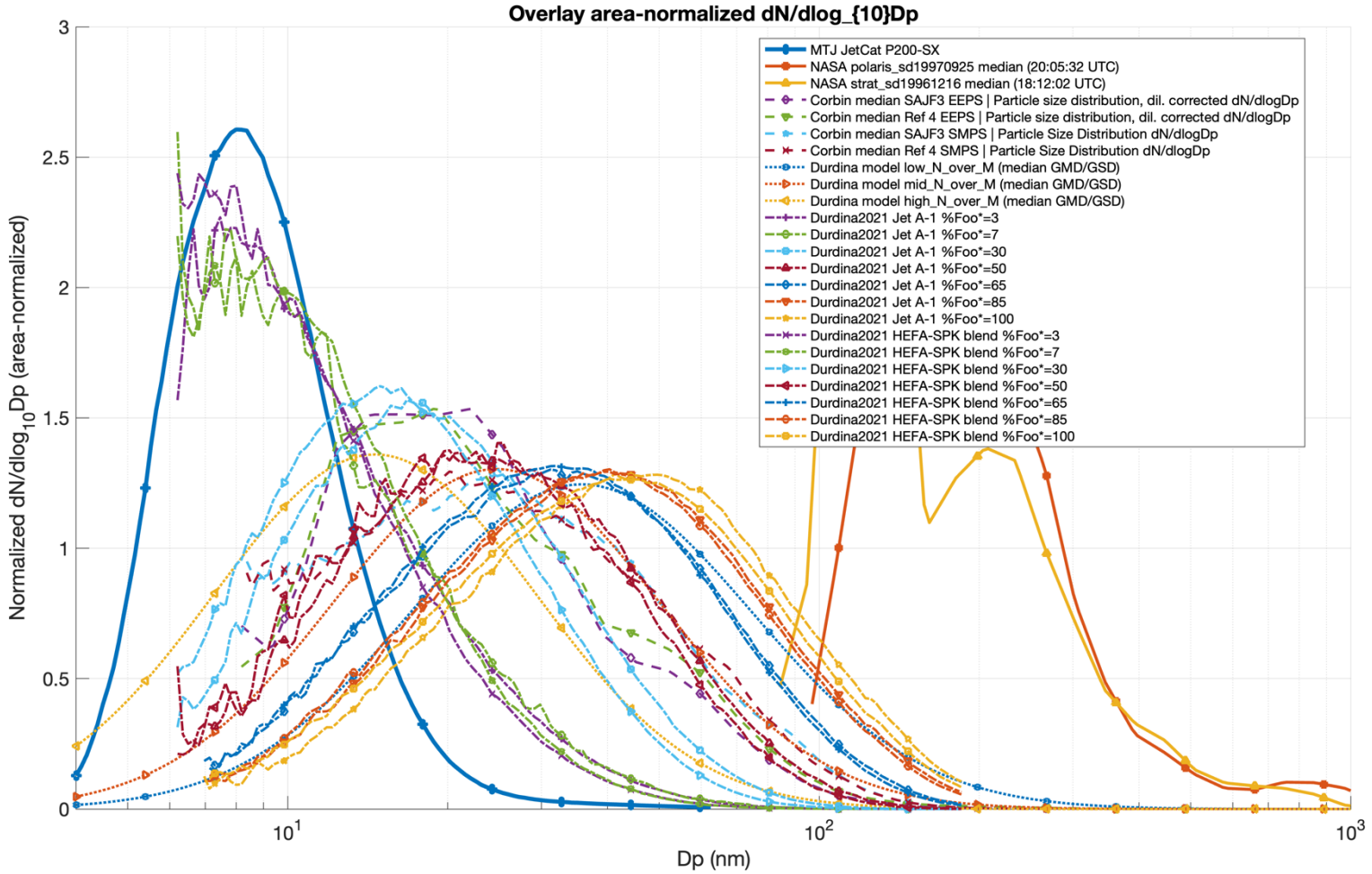


Instrument Uncertainties

Instrument	Species	Units	Accuracy
2B 405 nm	NO_x	PPB	±2 %
T300U	CO	PPM	±1 %
LI-850	CO_2	PPM	±1.5 %

RESULTS: MICRO-TURBOJET VS TURBOFAN (NVPM)

The measured distribution exhibits a distinct mono-modal peak with a mode diameter of 7.84 nm, **geometric mean diameter of 8.6** nm, and geometric standard deviation of 1.44.



[19] (TSI Inc. 2016), [21] (Durdina et al. 2021)

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